RUNNING TITLE: Training for performance and injury reduction

How Much? How Fast? How Soon? Three Simple Concepts for Progressing Training Loads to Minimize Injury Risk and Enhance Performance

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Abstract

Background: When progressing an athlete from rehabilitation to peak performance, load must exceed load-capacity. When gradual, systematic increases in load are applied, load-capacity will improve. However, if the applied load greatly exceeds load capacity, then tissue tolerance is exceeded, and injury may occur.

Clinical Question: It is well established that a balance exists between providing an adequate training stimulus to elicit performance benefits while also minimizing the risk of injury. How can practitioners determine how much training is too much? Following injury, how soon can training loads be progressed? How quickly can athletes return to competition?

Key Results: When developing rehabilitation or performance programs, three key concepts are critical: the "*floor*", the "*ceiling*" and *time*. The *floor* represents the athlete's current capacity, whereas the *ceiling* represents the capacity needed to perform the specific activities of the sport. A challenge in most sporting environments is the time required to progress from the *floor* to the *ceiling*. If athletes' training loads are progressed too rapidly, they will be at increased risk of injury and underperformance.

Clinical Application: Rehabilitation practitioners should consider and plan the appropriate amount of time required to progress from the *floor* (e.g. rehabilitation) to the *ceiling* (e.g. return to performance). The resilience and robustness that comes from training takes time, and different physical capacities will adapt at different rates. Progressive, gradual, and systematic increases in training load allows athletes to safely progress to the *ceiling*, reducing injury risk, improving availability, and enhancing performance.

Key Words: workload, injury, injury prevention, rehabilitation, load, training, strength and conditioning

Key Points

Findings: This clinical commentary addresses key considerations for rehabilitation practitioners when helping an athlete prepare for the demands of competition. Rehabilitation and reconditioning programs should prepare athletes for the most demanding passages of competition. This will ensure that (1) athletes are able to perform the high-intensity tasks that often determine the outcome of the game, and (2) athletes are at lower injury risk when performing these activities.

Implications: These findings highlight the importance of maintaining an adequate training load (or *floor*) during the off-season and while injured. Preparation for sporting activities involves year-round management, monitoring, and manipulation of training load, with an understanding that the *ceiling* is somewhat of a 'moving target' - different capacities will require development depending on the specific phase of the season, and the factors limiting performance for individual athletes.

Caution: Very young and older athletes, and those with a long injury history, poor training history, musculoskeletal deficiencies, lower strength and aerobic fitness may have poorer tolerance to rapid increases in training load, while other biomechanical and psychosocial factors can also impact load capacity. Along with gradual loading progressions, a

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comprehensive assessment of the sport-specific (e.g. strength, aerobic fitness) and tissue-specific (e.g. musculoskeletal) capacities is required to ensure training loads are progressed on an individual basis.

Data Sharing. There are no data in this manuscript

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Introduction: Athletes Need to Load in Order to Withstand Load!

Overload is a key principle of training – load must exceed capacity to improve performance. Small, systematic increases in load that are slightly greater than load capacity will improve tolerance to further load. However, if the applied load greatly exceeds load capacity, then tissue tolerance is exceeded, and injury may occur.[2,14] This clinical commentary addresses key considerations for rehabilitation practitioners when helping an athlete prepare for the demands of competition.

Clinical Questions: How Much? How Fast? How Soon?

Sports medicine practitioners play a role in maximizing the positive and minimizing the negative adaptations to training – too much training may result in excessive fatigue; too little training may mean athletes are underprepared for competition. Clearly a balance exists between providing an adequate training stimulus to elicit performance benefits while minimizing the risk of injury. How can practitioners determine how much training is too much? Following injury, how soon can training loads be progressed? How quickly can athletes return to competition?

The Floor, the Ceiling, and Time

When developing rehabilitation or performance programs, three key concepts are critical: the "*floor*", the "*ceiling*" and *time*. The *floor* represents the athlete's current capacity, whereas the *ceiling* represents the capacity needed to perform the specific activities of the sport. It is possible to safely progress an athlete from the *floor* to the *ceiling* as long as the athlete is afforded adequate *time* (Figure 1A).

The Challenges of Getting Pre-Season Right

A challenge in most sporting environments is the time required to progress from the *floor* to the *ceiling*. If athletes' training loads are progressed too rapidly, they will be at increased risk of injury.[5] This scenario occurs all too frequently. Consider athletes who enter a very short training camp (or pre-season period), or who have sustained an injury. If the gap between the current capacity and the required capacity is large, then the only way to progress from the *floor* to the *ceiling* is to rapidly increase training load to ensure that the athletes are prepared for the first competition game or return to play. Notwithstanding the fact that athletes with poorer physical capacities[9,11] and musculoskeletal dysfunction[12] are at increased risk of injury, training in this manner is associated with a high risk of injury[5] and poor performance[8] (Figure 1B). Coaches can take more time to safely progress to higher training loads and prepare athletes for the *ceiling*, but coaches do not have infinite time. Equally, if an inadequate training stimulus is applied, then the athlete is at risk of being underprepared, underperforming, and re-injury.

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How Does One Help Athletes Safely Progress from the "Floor" to the "Ceiling"?

The demands of elite performance are constantly evolving; in general, the complexity and physicality are increasing every year.[1] Reducing the *ceiling* is not a realistic option to help athletes avoid injury and perform well. One option is to take more time to bridge the gap between the *floor* and the *ceiling*, although most coaches will be less than impressed if their best athletes are not fit enough to compete in the first game of the season or spend extended periods of time in rehabilitation (Figure 1C). What if we could "buy" more time prior to the official start of the pre-season? If athletes performed a minimum volume of training prior to returning from an extended break, this would have the effect of artificially increasing the length of the pre-season period, minimizing the detraining effect induced by off-season, thereby ensuring that progression to the *ceiling* was gradual and systematic (Figure 1D).

Progressing athletes, of any performance level, from the *floor* to the *ceiling* is further complicated when the athlete returns to pre-season training in a severely deconditioned state or following off-season surgery. In these examples, the athlete's current capacity is inadequate to sustain normal training loads; rather than having an adequate *floor* their capacity is more like "*the basement*"! (Figure 1E). This situation can also occur when athletes are injured; local tissue capacity can immediately decrease, resulting in a reduction in the *floor*. In this respect, given the same amount of time, progressing the athlete from their current capacity to the *ceiling* would require rapid changes in training load – in turn, increasing injury risk.

Another solution to ensure that athletes are safely progressed from the *floor* to the *ceiling* is to raise the *floor* (Figure 1F) – ensuring that when athletes are taking an extended break or enter rehabilitation, they do not allow their physical capacity to fall to the *basement*. The benefit of raising the *floor* is that it also provides athletes the opportunity to develop greater load capacity than previously may have been possible (perhaps reaching "*the penthouse"!)* (Figure 1F). Preparation for sporting activities involves year-round management, monitoring, and manipulation of training load, with an understanding that the *ceiling* is somewhat of a 'moving target'[4] - different capacities will require development depending on the specific phase of the season, and the factors limiting performance for individual athletes.

Insert Figure 1 About Here

What Can Rehabilitation and Performance Staff Do to Help Athletes Achieve Their Performance Goals?

Athletes participate in sport for many reasons, which may differ between elite and non-elite competitors. Whether competing at an elite or non-elite level, athletes strive for continual improvement and to achieve their personal best performances. To achieve high-level performances, the load capacity of athletes must be adequate to meet the demands required of competition. Athletes cannot perform if their *current* capacity is well below the capacity *required* of their sport. Equally, athletes cannot perform if they are injured.

Taking the concepts of the *floor*, *ceiling*, and *time*, there are at least 5 simple ways rehabilitation and performance staff can minimize the risk of injury and give athletes the best chance of achieving their performance goals (summary box).

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Summary Box. Five Ways to Ensure Athletes Are Well Prepared for the Demands of Competition.

1. Maintain an adequate training load during the off-season[3] and while injured. Loading during these periods raises the *floor* (or alternatively ensures athletes avoid the *basement*) and improves the athletes' ability to tolerate load during pre-season and on return to competition.

2. Identify the *ceiling* and ensure that training load is proportionate to competition demands.[3] Various methodologies (including the use of wearable and video technologies) have been used to assess the sport-specific demands required at the *ceiling*[4,8,9,15]. These may include (but not necessarily limited to) peak running intensities[4], longest ball-in-play periods[8], repeated-sprint[15] and repeated-effort activity[9]. If expensive technology is not available, practitioners are encouraged to access sport-specific literature to inform their training programs. The individual load capacities, and the absolute competition demands will differ between elite and non-elite, adult and adolescent, and male and female athletes; training demands should also reflect these differences.

3. Assess individual differences in training tolerance among athletes. Very young and older athletes[11], and those with a long injury history[4], poor training history[10], musculoskeletal deficiencies[12], lower strength[9] and aerobic fitness[11] may have poorer tolerance to rapid increases in training load. Furthermore, other physical (e.g. biomechanical, movement patterns) and psychosocial (e.g. emotional, lifestyle) factors can impact load capacity[6]. It might be tempting to rapidly increase training loads in less fit athletes and those with musculoskeletal deficiencies. However, these are the athletes who are least likely to tolerate this type of training progression, and in turn most likely to sustain injury in response to this loading pattern. Along with gradual loading progressions, a comprehensive assessment of the sport-specific (e.g. strength, aerobic fitness) and tissue-specific (e.g. musculoskeletal) capacities is required to ensure training loads are progressed on an individual basis. It is here that sports medicine professionals (e.g. physical therapists and strength and conditioning coaches) play a critical role.

4. Identify and prepare for the most demanding passages of play. Training for the average demands of competition may mean athletes are underprepared for the "worst-case scenario".[7,13] This may trigger at least 2 consequences: (1) athletes are unable to perform the high-intensity tasks that often determine the outcome of the game,[13] and (2) athletes are at greater injury risk when attempting to perform these activities.

5. Training programs require an understanding of the (1) physical demands of the sport, (2) physical capacities required to perform these activities, and (3) factors that limit performance on an individual basis. Coaches should consider and plan the appropriate amount of time required to progress from the *floor* to the *ceiling*. The resilience and robustness that comes from training takes time, and different physical capacities will adapt at different rates. Progressive, gradual, and systematic increases in training load allows athletes to safely progress to the *ceiling*, reducing injury risk, improving availability, and enhancing performance.

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Summary: From Risk to Resilience

Training loads can have positive and negative effects. The risk of injury increases with rapid changes in training load. However, not all training load is bad – when prescribed appropriately it can create resilient and robust athletes, capable of withstanding the high loads of competition and thriving in the most demanding passages of play. Systematically increased training loads, and identifying factors related to load tolerance, not only lowers the risk of injury but allows athletes to progress to higher training loads typical of those required for elite performances.[5] Importantly, once athletes have reached these high training loads, they are at reduced risk of injury, and have greater likelihood of achieving their performance goals – loading allows athletes to withstand further load!

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FIGURE LEGEND

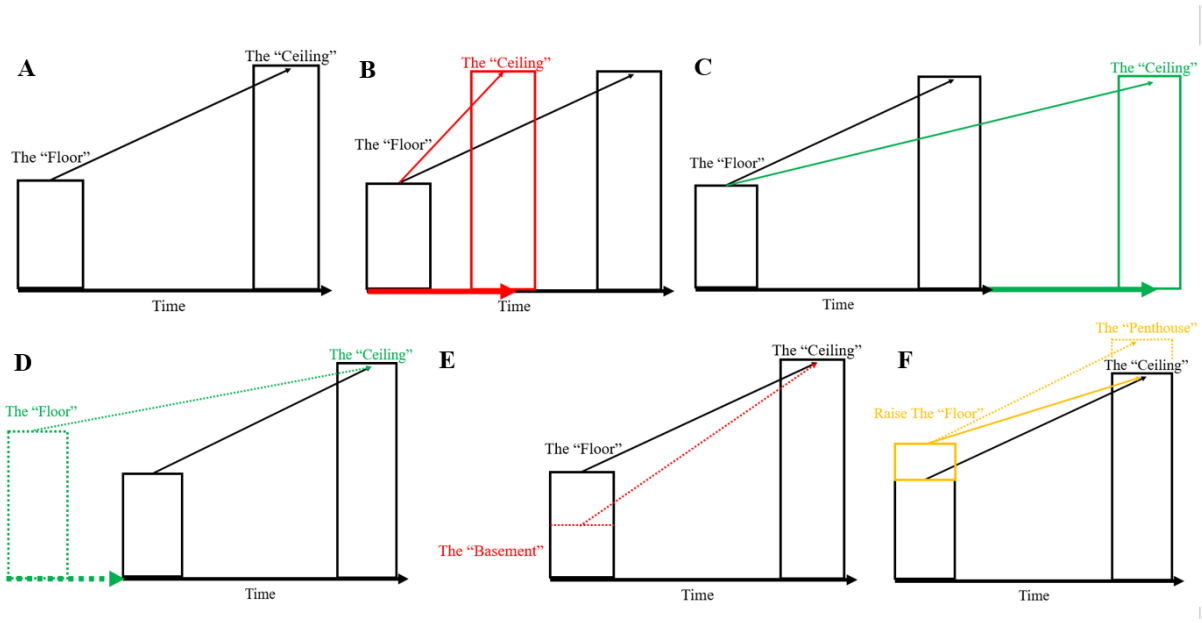

Figure 1. Different loading strategies that may be used to progress athletes from the "*floor*" (i.e. current capacity) to the "*ceiling*" (i.e. required capacity).

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